



Rain Drive



Our Team

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Problem Statement

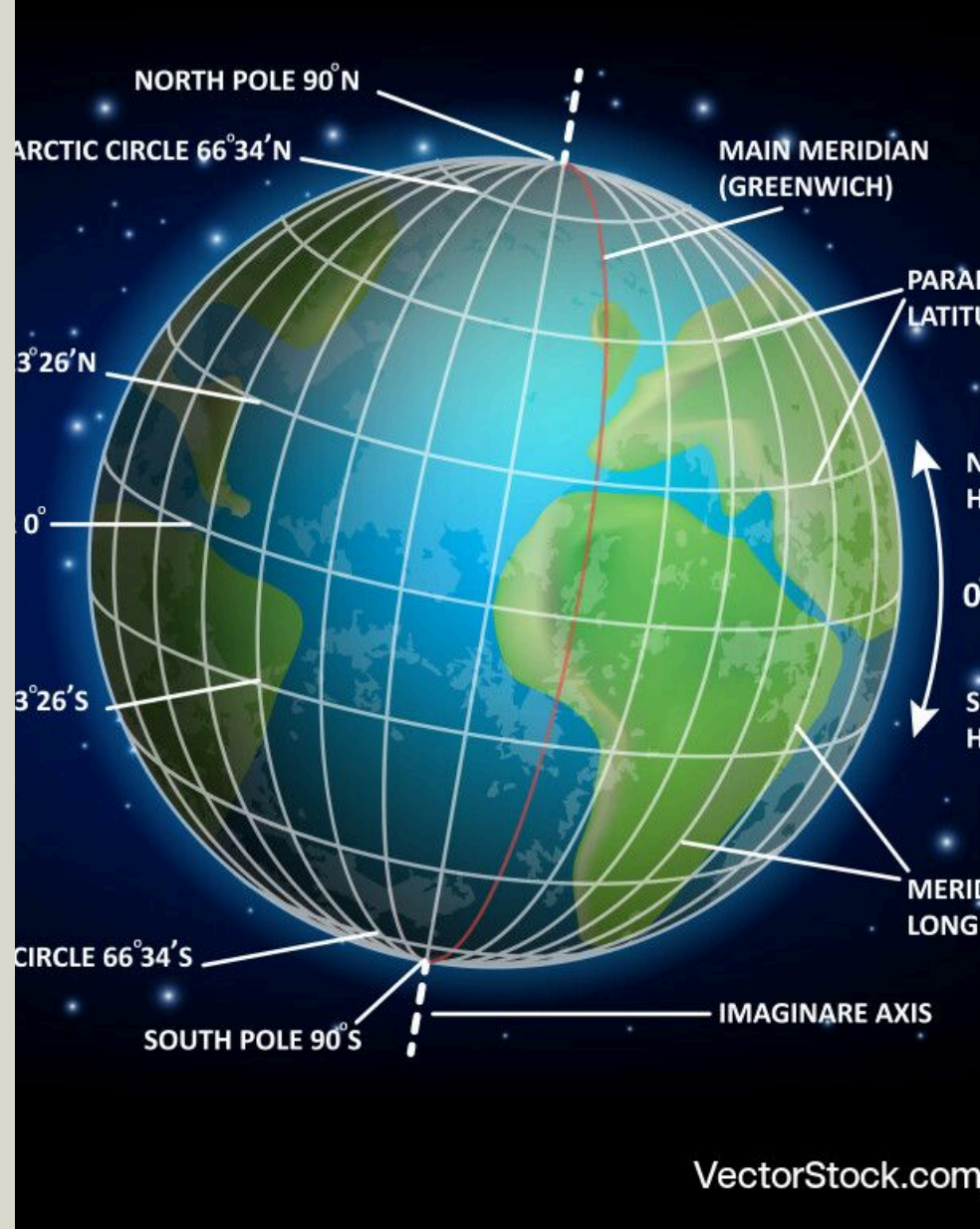
Will It Rain On
My Parade ?





What's the challenge all about??

If you're planning an outdoor event—like a vacation, a hike on a trail, or fishing on a lake—it would be good to know the chances of adverse weather for the time and location you are considering. There are many types of Earth observation data that can provide information on weather conditions for a particular location and day of the year. The challenge is to construct something with a personalized interface that enables users to conduct a customized query to tell them the likelihood of “very hot,” “very cold,” “very windy,” “very wet,” or “very uncomfortable” conditions for the location and time they specify.



Our Solution

We developed a Python program that:

Takes latitude and longitude as user input

Fetches real NASA weather data (temperature, wind, precipitation)

Detects extreme weather conditions based on set thresholds

Prints out results showing how comfortable or risky the day is





Data Source

✓ Dataset:

MERRA-2 (Modern-Era Retrospective Analysis for Research and Applications)

✓ Accessed via:

NASA GES DISC OPeNDAP

✓ Parameters Used:

T2M → Temperature (Kelvin)

U10M, V10M → Wind Components (m/s)

PRECTOT → Total Precipitation (mm)



How our code works?

- 1) User enters latitude and longitude.
- 2) The program connects to NASA's MERRA-2 dataset via OPeNDAP.
- 3) It finds the nearest grid point for that location.
- 4) Extracts temperature, wind, and rainfall data.
- 5) Converts temperature from Kelvin → Celsius.
- 6) Calculates wind speed using U and V components.
- 7) Checks for extreme conditions based on thresholds.
- 8) Prints out weather summary and detected conditions.

How Can We Help



Gives users a clear understanding of whether their day will be hot, cold, windy, or rainy.



Helps in planning outdoor activities safely.



Uses real NASA Earth observation data — ensuring scientific accuracy.



Step towards a personalized weather assistant